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Exp-6: Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Aim

To implement the Naïve Bayes algorithm in Python and evaluate its performance using a confusion matrix and accuracy score.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create a Gaussian Naïve Bayes classifier.
5. Train the model using the training data.
6. Predict the class labels for the test data.
7. Generate the confusion matrix.
8. Calculate the accuracy score.
9. Display the predicted labels, confusion matrix, and accuracy.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score

# Load the Iris dataset

iris = load_iris()

X = iris.data    # Features

y = iris.target  # Target labels

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the Naïve Bayes model

model = GaussianNB()

# Train the model

model.fit(X_train, y_train)
```

```

# Predict the test data
y_pred = model.predict(X_test)
# Display Actual and Predicted values
print("Actual Labels :", y_test)
print("Predicted Labels:", y_pred)
# Generate Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

```

OUTPUT

RESULT

```

... Actual Labels   : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
   Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

   Confusion Matrix:
   [[10  0  0]
    [ 0  9  0]
    [ 0  0 11]]

   Accuracy: 100.00%

```

RESULT

Thus, the Naïve Bayes algorithm was successfully implemented in Python, and the classification performance was evaluated using the confusion matrix and accuracy

